COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	07/22/26	**DATE SUBMITTED:**	07/22/26
NAME:	Hare, Andrei Angelo D.	**INSTRUCTOR:**	Engr. Jimlord Quejado

SEATWORK NO. 3.1 Singly Linked List

1. PROCEDURE OUTPUT

What procedures did you perform?

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Google Gemini, Quillbot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction, idea generation**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

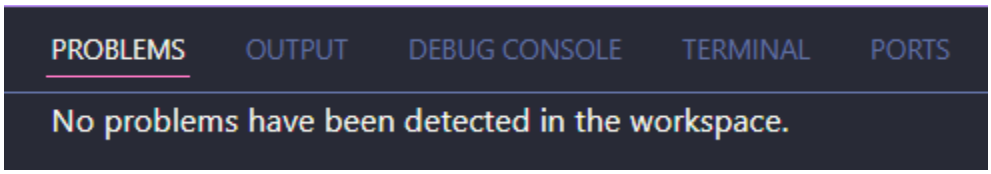
Extent of Use:

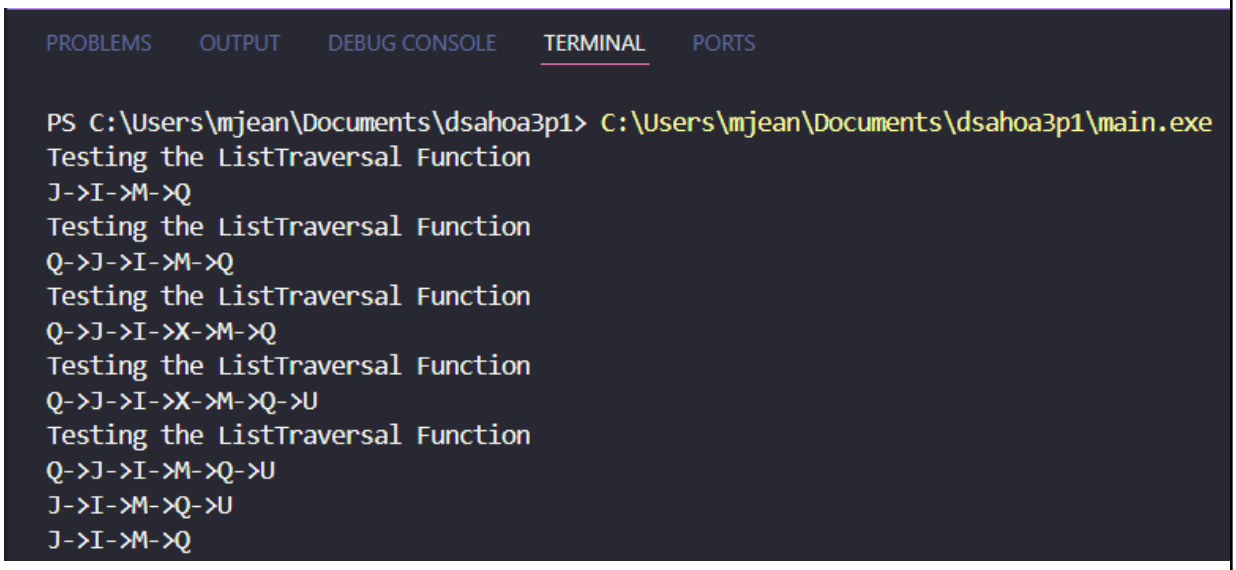
Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Output:





Code Analysis:

On this seatwork, I was able to observe how a singly linked list functions in C++. I discovered that every time a new node is added or removed, the list changes because each node stores data and points to the next node. I could see from the output that the list began with J, I, M, and Q and changed when I added a node at the head, another at the middle, one at the end, and removed some nodes. This made me think that linked lists are connected by pointers rather than being kept in one location like arrays.

Reflection:

This lesson was a little challenging for me because I just totally didn't recall what I had learned C++ and OOP because it's been awhile so I'm still adjusting and trying to catch-up. At first, I found it difficult to understand how pointers and linked lists work together but somehow after following the discussion and seeing how the code changes the list after every operation, I became quite familiar with the concept. I still need more practice to fully understand it, but this activity gave me a better idea of how singly linked lists work. I hope to improve my understanding as we continue studying the course.

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